3.3. CAPM LESSON 3: THE AVERAGE INVESTOR HOLDS THE MARKET

The market portfolio represents the average holdings across investors. The intersection of the CAL with the mean-variance frontier represents an investor who holds 100% in the MVE portfolio. This tangency point represents the average investor. The risk aversion corresponding to this 100% portfolio position is the risk aversion of the market.[3]

Note that as investors differ from the average investor, they will be exposed to more or less market factor risk depending on their own risk preferences. We will come back to this extensively in chapter 14 when we describe factor investing.

3.4. CAPM LESSON 4: THE FACTOR RISK PREMIUM HAS AN ECONOMIC STORY

The CAL in Figure 6.3 for a single investor is called the capital market line (CML) in equilibrium, since under the strong assumptions of the CAPM every investor has the same CML. (The MVE portfolio is the market factor portfolio.) The equation for the CML pins down the risk premium of the market:

$$E(r_m) - r_f = \bar{\gamma}\sigma_m^2, \tag{6.1}$$

where $E(r_m) - r_f$ is the market risk premium, or the expected return on the market in excess of the risk-free rate; $\bar{\gamma}$ is the risk aversion of the "average" investor; and σ_m is the volatility of the market portfolio. The CAPM derives the risk premium in terms of underlying agent preferences ($\bar{\gamma}$ is the average risk aversion across all investors, where the average is taken weighting each individual's degree of risk aversion in proportion to the wealth of that agent).

According to the CAPM in equation (6.1), as the market becomes more volatile, the expected return of the market increases and equity prices contemporaneously fall, all else equal. We experienced this in 2008 and 2009 when volatility skyrocketed and equity prices nosedived. Expected returns in this period on were very high (and realized returns were indeed high in 2009 and 2010). It is intuitive that the market risk premium in equation (6.1) is proportional to market variance because under the CAPM investors have mean-variance preferences: they dislike variances and like expected returns. The market portfolio is the portfolio that has the lowest volatility among all portfolios that share the same mean as the market, or the market has the highest reward-to-risk ratio (or Sharpe ratio). The market removes all idiosyncratic risk. This remaining risk has to be rewarded, and equation (6.1) states a precise equation for the risk premium of the market.

[3] Technically speaking, the market is the wealth-weighted average across all investors.